

PULSE-Cat: Catalyst-Referenced Pose Learning and Progressive Substrate Selection in Small-Data Catalysis

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S1 | Study scope and guide to supporting information

This Supporting Information describes the datasets, molecular representations, validation protocols, and retrospective analyses underlying PULSE-Cat. PULSE-Pose supplies catalyst-referenced substrate representations; PULSE-Acquire evaluates sequential substrate selection; PULSE-Transfer examines guarded cross-catalyst learning; and PULSE-Explore compares candidate generation and enumeration. Feature identifiers are retained to distinguish the representation and validation experiments.

The study is retrospective and prediction-based. The reported experimental outcomes originate from the two literature reaction datasets, while computational poses and generated candidates do not constitute additional experimental observations. Section S9 describes the companion SI .zip data/code archive, which will be supplied as a separate supplementary file at submission. Its directory map follows the release inventory dated 9 September 2026. Protocol-specific populations, the distinction between documented transfer suppression and contract-level safeguards, and the incomplete enumeration scoring comparison are stated explicitly below.

The primary numerical endpoint is unsigned enantiomeric excess (ee), in percent; its MAE and RMSE are in ee percentage points. Isolated yield is a separate endpoint. No generated geometry, predicted candidate, or unreported experiment is counted as an additional measured substrate. Literature citations support the underlying methods and chemical sources, not the numerical results of this study.

TABLE S1 | Support map for the current main manuscript. Section and table numbers refer to this Supporting Information.

<i>Main manuscript topic</i>	<i>SI location and interpretation</i>
Methods and dataset scope	Sections S2–S4 cohort definitions are not interchangeable.
Results 3.1: connectivity and compact poses	Table S3 and Tables S13–S14 these are fixed-replay feature-arm comparisons.
Results 3.1: matched pose gain, occupancy, Stage 3	Table S5 each comparison retains its own validation protocol.
Results 3.1: CREST/GFN2-xTB	Section S5.2, Tables S6 and S7 logs and coordinates are indexed in Section S9.4.
Results 3.1: 91 density fields and 13 overlays	Section S3.4 fields and figures are indexed in Section S9.4.
Results 3.2: 55.3% and 33.2% reductions	Table S8 mean-prefix errors, not prospective experimental savings.
Results 3.2: estimator, n_{90} , and yield exceptions	Tables S9–S12 final-prefix pools depend on route.
Results 3.3: transfer and controls	Section S7 fixed replay, randomized routes, and scaffold bootstraps are separated.
Results 3.4: generation and enumeration	Section S8 panel records and the scoring exclusion are detailed in Section S9.5.
Abstract, Discussion, and Conclusion	Same protocol-matched evidence Section S9 limits mechanistic, transfer, and prospective claims.

S2 | Datasets, endpoints, and distinct analysis populations

S2.1 | Reaction sources and curation

The experimental domains are Fe(P7)Cl and Fe(CMC-Por)Cl intramolecular C(sp³)-H amination, as reported independently by Lee and co-workers and Wang and co-workers, respectively [1, 2]. The curation specification records reaction and substrate identifiers, catalyst and condition identifiers, starting-material azide SMILES, free-nitrene SMILES, product SMILES when reported, reactive-site atom maps, experimental ee/yield, family and scaffold groups, exclusions, source order, and provenance. Missing product structures remain null rather than being computationally reconstructed. Missing ee is not zero and is not replaced by yield. A measured zero-ee outcome remains a valid observation; an unreported reaction is not assigned a failure label.

Canonical matching removes atom-map labels before RDKit canonicalization [3]. The 13 canonical aryl substrates shared between the domains are identified in the overlap ledger. They may be used for paired diagnostics, but their source counterparts are excluded from cold-start transfer training. The release inventory locates the row-level ledgers and source links under `curated-datasets/`; see Section S9.1.

The independent progression summaries end at 37 revealed Fe(P7)Cl ee labels, 38 yield labels, and 25 Fe(CMC-Por)Cl aryl labels, leaving a route-dependent residual pool. A separate reciprocal guide reports 37 scored Fe(P7)Cl and 31 scored Fe(CMC-Por)Cl events, and refers to a 40-record Fe(P7)Cl target ledger. These event counts must not be substituted for the 38-substrate development cohort. Removing only 1ad from 41 records cannot explain a 39-record progression set. The release inventory identifies `dataset-id-lists.csv` as the explicit reconciliation; its population memberships and exclusion reasons must be used when reproducing each analysis.

TABLE S2 | Reported populations. Counts describe different analysis inputs, not successive interchangeable versions of one dataset.

<i>Analysis population</i>	<i>Count</i>	<i>Definition and qualification</i>
Fe(P7)Cl retained ledger	41	Substrate-identity records; includes the valid zero-ee record 1z.
Fe(P7)Cl matched development ee	38	17 families; excludes 1ad, 1an, and 1z. Reported cohort identifier f11a0a2ec491.
Initial Fe(P7)Cl yield benchmark	41	19 families; distinct from progressive yield.
Fe(P7)Cl CREST completion set	40	Excludes 1z; removing 1ad and 1an leaves all 38 modeled substrates.
Fe(P7)Cl independent progression	38/39	38 usable ee and 39 yield labels; the release inventory identifies dataset-id-lists.csv for reconciliation.
Fe(CMC-Por)Cl retained ledger	50	33 aryl and 17 sulfonyl substrates; sulfonyl substrates are a pathway control.
Fe(CMC-Por)Cl independent progression	26	Active aryl labels per endpoint in the independent progression record; not the complete 33-record aryl ledger.
Shared canonical aryl identities	13	Cross-catalyst overlap; excluded from source training as specified above.
Density-field inventory	91	41 Fe(P7)Cl plus 50 Fe(CMC-Por)Cl substrate-level fields, not 91 unique identities across domains.

S3 | Molecular features and catalyst-referenced PULSE-Pose

S3.1 | Molecular connectivity and the reciprocal feature ladder

Molecular inputs are sanitized starting-material azide SMILES. Missing or unparsable SMILES and duplicate substrate identifiers stop feature-table construction. Outcome, catalyst, source-order, and grouping metadata are not molecular predictor columns. Morgan bit vectors follow circular-fingerprint methodology [4]. B3–B5 use radius 2, binary rather than count fingerprints, and `includeChirality=False`.

The 15 RDKit descriptors, in the documented column order, are heavy-atom count, molecular weight, Crippen logP [5], topological polar surface area [6], formal charge, fraction of sp^3 carbon, rotatable-bond count, total ring count, aromatic-ring count, hydrogen-bond acceptor count, hydrogen-bond donor count, heteroatom count, aliphatic-ring count, spiro-atom count, and tetrahedral chiral-centre count. The last counts assigned and unassigned centres through `Chem.FindMolChiralCenters(includeUnassigned=True)`; it is not an absolute-configuration encoding. B1 retains molecular weight, logP, polar surface area, fraction sp^3 , rotatable bonds, and total rings.

TABLE S3 | Reciprocal feature ladder. Dimensions precede training-fold filtering.

Arm	Definition	Columns
B0	Training-fold mean	0
B1	Six RDKit descriptors	6
B2	Fifteen RDKit descriptors	15
B3	B2 + 128-bit Morgan	143
B4	B2 + 256-bit Morgan	271
B5	B2 + 1,024-bit Morgan	1,039
P0	Stage 2p catalyst-referenced fields	248
P1	B1 + P0	254
P2	B5 + all P0 features	1,287
P3	B5 + compact C1 geometry	1,083
P4	B5 + C1 + C2 catalyst interactions	1,099

C1 contains 44 retained geometry summaries; C2 contains 16 explicit catalyst-interaction summaries. Their functional categories include attack-corridor geometry, directed fields, stratum contrasts/stability, catalyst contacts, pocket crowding, and corridor obstruction. Exact selected column names are referred to the released feature manifests, data dictionary, and feature source described in Section S9.4. Repeated numeric interaction values are averaged by substrate before selecting C2; missing required interactions cause failure, not zero imputation. A separate legacy candidate-site builder uses a chirality-aware 2,048-bit fingerprint and is not B5. Literature on cavity descriptors motivates catalyst-aware representations but does not validate this study's feature selection [7, 8].

S3.2 | Label-blind embedding, rigid assembly, and selection

The canonical Stage 1 pose generator starts from free-nitrene SMILES with explicit hydrogens, unlike the azide input to B5 and CREST. It requests 5,000 ETKDGv3 conformers with seed 20260730 and performs UFF relaxation with a 500-iteration limit per call [9–11]. The nitrene nitrogen and its bonded neighbour align the substrate to corresponding reference atoms. Catalyst-core coordinates are copied unchanged and the reference Fe–N distance is checked to 10^{-5} Å. This is rigid catalyst-referenced assembly and screening, not UFF optimization of the complete catalyst–substrate complex.

Candidate acceptance requires zero optimization status, preserved canonical connectivity, and no severe catalyst–substrate clash. Except for the intended Fe–nitrene-N pair, a separation below $0.65(r_a^{\text{cov}} + r_b^{\text{cov}})$ fails the clash screen. The documented covalent radii in Å are H 0.31, B 0.85, C 0.76, N 0.71, O 0.66, F 0.57, Si 1.11, P 1.07, S 1.05, Cl 1.02, Br 1.20, I 1.39, and Fe 1.32. Covalent-radius conventions are described by Cordero and co-workers [12]; the numerical lookup and clash factor above define this study's screen, not a transferable force-field potential. Valid candidates are ordered by UFF score and conformer ID. The lowest 100 and highest 100 non-overlapping valid candidates are retained; fewer than 200

cause failure rather than duplication or padding. An additional cleanup call at writing records final geometry, energy, connectivity, and core checks; pose acceptance reports and run logs are indexed in Section S9.4.

No ee or yield is used in generating or selecting poses. Nevertheless, the downstream reaction frame depends on the reported reactive-site atom map. It is therefore a site-conditioned retrospective representation, not evidence of prospective site discovery. Low/high UFF strata are geometric ranks, not free energies, transition states, or Boltzmann populations, and 200 retained poses need not be 200 distinct conformational basins.

S3.3 | Stage 2p fields and compact catalyst information

Let N, H, and C denote the nitrene nitrogen, transferred hydrogen, and reported reactive carbon. The forward axis is $\hat{\mathbf{u}} = (\mathbf{H} - \mathbf{N})/\|\mathbf{H} - \mathbf{N}\|$ with origin N; the reverse view has origin H and direction $-\hat{\mathbf{u}}$. For atom position $\mathbf{x}$ relative to origin $\mathbf{o}$,

$$z = (\mathbf{x} - \mathbf{o}) \cdot \hat{\mathbf{u}}, \quad r = \|\mathbf{x} - \mathbf{o}\|, \quad (\text{S1})$$

$$\rho = \sqrt{r^2 - z^2}, \quad \cos \theta = z/r. \quad (\text{S2})$$

Zero radial distance receives cosine zero; coincident anchors raise an error. Azimuth is discarded. Catalyst atoms and the anchor nitrene N are excluded from substrate-heavy-atom fields; catalyst context enters via the assembly and screening, whereas C2 contains explicit interaction summaries.

Nitrogen-centred channels cover all heavy atoms, carbon, heteroatoms, aromatic atoms, and halogens. Hydrogen-centred channels also include the reactive hydrogen. Channel quantities are mass relative to substrate-heavy-atom count, mean and 90th-percentile radius, near-field response, and even/odd angular norms from mean Legendre moments of orders (2, 4) and (1, 3). Radial kernels are $\exp[-(r - \mu_j)^2/(2\sigma_j^2)]$, with centres (1, 2, 3, 4, 5.5, 7, 9) Å and widths (0.70, 0.70, 0.75, 0.80, 0.95, 1.10, 1.30) Å; the near response averages the first three. Empty channels return zero.

Directed all-heavy and heteroatom channels describe mean/90th-percentile axial position and occupancy of $0 \leq z \leq \|\mathbf{H} - \mathbf{N}\|$, $\rho \leq 2.5$ Å. Features are averaged over low-UFF, high-UFF, and combined poses. The 248 columns comprise 198 field means, 36 directed means, four symmetry/antisymmetry summaries, and ten relational features. Relational features compare radial distributions using Wasserstein/energy distances, angular histograms using Jensen–Shannon divergence, and radial means with UFF rank. Standard definitions of the distribution distances are documented in the SciPy reference manual [13, 14]. Jensen–Shannon divergence must be distinguished from its square-root distance; the histogram normalization, logarithm base, and implemented form are specified by the feature source. Exact file-level field ordering still requires the feature manifest.

S3.4 | Occupancy, flexibility, and pose-density visualizations

Stage 2pp produces 655 occupancy/flexibility features using the site-conditioned frame. Its cylindrical grid has 40 axial bins from -8 to 12 Å and 20 radial bins from 0 to 10 Å, at 0.5 Å spacing. Nitrene-centred, carbon-centred, N-to-H, and H-to-N views use all-heavy, carbon, heteroatom, halogen, hydrogen, reactive-carbon, and reactive-hydrogen channels. Atom kernels in (z, ρ) have $\sigma_a = \max(0.22, 0.30r_a^{\text{vdW}})$ Å and are truncated at three widths. Van der Waals radii in Å are H 1.20, B 1.92, C 1.70, N 1.55, O 1.52, F 1.47, Si 2.10, P 1.80, S 1.80, Cl 1.75, Br 1.85, and I 1.98; other elements use 1.70. Bondi and subsequent main-group tabulations provide the van der Waals radius conventions [15, 16]; the listed values, fallback, and kernel-width rule are the explicit computational choices for this representation.

Duplicate-like geometries are balanced within each stratum using a signature of N–H and N–C distances, the N–C–H angle, and the first 40 upper-triangle heavy-atom pair distances. Each component is divided by 0.35 and rounded. This mixed-component quantization is not a symmetry-aware RMSD cutoff. Signature clusters receive equal weight, shared among their members; combined fields assign half the weight to each UFF stratum. This differs from unweighted Stage 2p means and from Boltzmann weighting.

The archive reports 91 substrate NPZ fields and 91 SVG views, plus paired overlays for 13 shared identities. Each has low-UFF, high-UFF, combined, and difference representations. The release inventory locates these files in the density directories and overlay index described in Section S9.4. The field counts describe the documented geometry inventory; they do not establish distinct conformational basins or mechanistic assignments. Density means geometric occupancy, not electron density or a selectivity-determining interaction.

S4 | Model fitting, validation, and error definitions

S4.1 | Leakage controls and protocol separation

A substrate, not a pose, is the statistical observation. All of its poses remain in the same split. Standardization, constant-feature filtering, and any learned feature selection or projection must use training data only, separating model selection from performance evaluation [17]. Family and scaffold exclusions are separate tests; ordinary sequential replay is not relabelled as a scaffold-held-out evaluation.

In target-held-out development, one of the 38 Fe(P7)Cl substrates is withheld. MaxMin Morgan/Tanimoto selection chooses 29 training substrates from the other 37; eight form a validation set for model selection. For held-out families of size 1, 2, 3, or 12, documented training/validation counts are respectively 30/7, 29/7, 28/7, and 21/5. The release inventory identifies the model-report split files for family assignments and split membership (Section S9.2). The matched structure-only and structure-plus-pose matrices contain 547 and 1,241 columns: they are not B5 and P2. Their ordered feature lists, selected parameters, and fitting settings should be matched to the released feature/report trees and source code, rather than inferred from dimensions alone.

Later Stage 3 and CREST comparisons use the same 38 substrates from 17 families with outer and inner leave-one-family-out evaluation, but a different validation implementation and locked 248-column geometry reference. The separate occupancy comparison uses its own reported local-geometry reference. No row in Table S5 should be compared across protocol blocks as if it were a paired ablation.

The estimator families include Ridge regression [18], Elastic Net [19], LightGBM [20], and Gaussian-process regression [21]. Scikit-learn provides the software framework for the documented conventional estimators and preprocessing [22]. These references establish the methods and software, not the study-specific hyperparameters, selection procedures, or performance; Table S4 records the available fitting settings.

The documented nested implementation suppresses convergence warnings; suppression is not evidence that every fit converged. Progressive bounded predictions must not be substituted for the unbounded predictions in the nested descriptor ablation.

S4.2 | Prediction metrics and uncertainty

For held-out observations y_i and predictions $\hat{y}_i$,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|, \quad (\text{S3})$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}, \quad (\text{S4})$$

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (\text{S5})$$

$\bar{y}$ is the evaluation-cohort mean. A negative R^2 indicates more squared error than this reference. Percentage MAE reduction is $100(M_{\text{reference}} - M_{\text{candidate}})/M_{\text{reference}}$. Recomputing a percentage from rounded aggregate MAEs checks arithmetic only; it does not reproduce the model.

Bootstrap resampling estimates sampling variability by drawing observations with replacement [23]. Confidence intervals are limited to those documented for the corresponding analysis. CREST paired bootstraps resample substrate IDs with replacement using identical indices for both models, 20,000 draws, and NumPy seed 20260907. The statistic is baseline MAE minus augmentation MAE, so positive values favour augmentation. This is not a family-cluster bootstrap, a posterior probability, or prospective experimental confidence. Transfer route/scaffold uncertainty uses different averaging units, discussed in Section S7.

TABLE S4 | Parameters explicitly recoverable from the documented methods. Implementation details are indexed in Section S9.3; unspecified values are not filled with assumed defaults.

<i>Model or analysis</i>	<i>Recorded settings and limitations</i>
Progression/transfer Ridge	Standardized Ridge, $\alpha = 10$; predictions clipped to $[0, 100]$.
Independent progression Elastic Net	$\alpha = 0.1$, ℓ_1 ratio 0.15, maximum 10,000 iterations, random state 0.
Progression Tanimoto- k NN	Morgan-fingerprint similarity; k , weighting, and tie conventions are referred to <code>algorithms/ALGORITHM_SPECIFICATIONS.md</code> and the released source.
Nested geometry/CREST Elastic Net	$\alpha \in \{0.01, 0.1, 1, 10, 100\}$; ℓ_1 ratio $\in \{0.2, 0.5, 0.8, 1.0\}$. Inner pooled substrate MAE selects parameters; ties use ascending α , then ascending ratio.
Nested fit implementation	Maximum 5,000 iterations, tolerance 10^{-3} , random coordinate selection, random state 0. Training-zero-variance columns removed and training-only standardization applied; training-mean fallback if none remain. Predictions not clipped.
Other development estimators	Ridge, Elastic Net, LightGBM, and Gaussian process were examined; their fitting settings are referred to the released scripts, configuration, and model-report manifests.

S5 | Fe(P7)Cl representation development and isolated-substrate calculations

S5.1 | Protocol-matched development results

TABLE S5 | Fe(P7)Cl ee development on the matched 38-substrate cohort. Each block is a separate validation experiment; all rows are Elastic Net results.

Protocol and representation	MAE	RMSE	R^2
Target holdout: structure only	12.612	15.183	0.064
Target holdout: structure + pose	8.583	11.932	0.422
Family holdout: structure only	10.406	13.525	0.257
Family holdout: structure + pose	9.334	12.557	0.359
Occupancy comparison: local geometry	10.281	13.023	0.311
Occupancy comparison: +655 features	14.132	17.005	-0.175
Nested families: locked geometry	8.498	11.357	0.476
Nested families: +Stage 3a-alt	8.468	11.289	0.482
Nested families: +Stage 3b	8.664	11.978	0.417
Nested families: +Stage 3a-alt +Stage 3b	8.411	11.810	0.433

The target- and family-holdout pose gains are 4.029 and 1.072 ee points, or 31.9% and 10.3%, respectively. The occupancy augmentation worsens MAE by 3.851 points (37.5%). Stage 3a-alt changes MAE by only -0.030 points; the combined Stage 3 block lowers MAE by 0.087 but worsens RMSE and R^2 . No paired uncertainty interval for these small Stage 3 changes is available. These comparisons support selective retention of information, not a universal advantage of larger representations. The published descriptor literature provides methodological context, not evidence for these specific gains [7, 24].

Stage 3a-alt is described as a 21-feature hetero-site-competition block. A separate 40-row, 567-feature source matrix and a later Stage 2p join appear in the documented account. Their exact column-level mapping must be checked in the released feature files and manifests; summary dimensions alone do not specify this mapping. Stage 3b supplies intrinsic C-H bond-dissociation and within-substrate competition quantities. The documented calculation uses GFN2-xTB/ALPB ether neutral and radical optimizations, UHF 1 for the radical, a reported H-atom reference of -0.391629889244 Hartree, and

$$D_{\text{C-H}} = 627.509474(E_{\text{radical}} + E_{\text{H}} - E_{\text{neutral}}) \quad (\text{S6})$$

in kcal mol⁻¹ for energies in Hartree. These quantities are not catalyst-bound activation barriers. The separate restrained catalyst-bound Stage 3a pilot and rule/BDE calibration branch are not substituted for the Stage 3a-alt ablation reported in the main manuscript.

S5.2 | CREST/GFN2-xTB scope and execution

This comparator uses isolated neutral aryl-azide substrates with explicit hydrogens: no porphyrin, Fe, chloride, catalyst substituent, or frozen-atom constraint is present. Charge 0 and UHF 0 specify the closed-shell calculation. CREST provides automated conformational exploration [25]; the electronic-structure method is GFN2-xTB [26]. The documented logs identify CREST 3.0.2 and xTB 6.7.1, commit edcfbbe.

The initial RDKit ETKDGv3 seed is the first seven hexadecimal characters of SHA-256 of the UTF-8 string <domain>: <substrate_id>, interpreted in base 16. All 40 completed Fe(P7)Cl substrates used MMFF94 relaxation [27]; the UFF fallback was not used. The cleanup return code was not checked by the documented generator, so later xTB convergence does not independently verify that cleanup. Catalyst-bound Stage 1 XYZ coordinates did not initialize CREST, and later placement back against a porphyrin was not performed.

Production used four concurrent substrate workers, one thread each, a 1,800-s search timeout, and a target of ten retained conformers. The calculation command forms are shown below; structures and logs are indexed in the computational release, with dependencies and runtime records described in Section S9.6.

```
.chem-env/bin/crest input.xyz --gfn2 --alpb ether \
  --chrg 0 --uhf 0 --quick --legacy --nocross \
  --ewin 30 -T 1 --xnam .chem-env/bin/xtb
.chem-env/bin/xtb conformer.xyz --gfn 2 --alpb ether \
  --chrg 0 --uhf 0
.chem-env/bin/xtb conformer.xyz --gfn 2 --alpb ether \
  --chrg 0 --uhf 0 --json
```

The search uses reduced-cost legacy sampling, ALPB implicit solvation with ether parameters [28], disabled crossing, and a 30 kcal mol⁻¹ energy window. Logged CREGEN thresholds were RMSD 0.1250, rotational-constant/Bconst 0.0100, and population 0.0500; units or normalizations not specified in the calculation record remain unspecified. Independent reranks require exit code zero and a parseable total energy, with OMP_NUM_THREADS=1 and a 1,800-s timeout. JSON electronic single points use a 300-s timeout, without geometry optimization.

The calculation record reports 40/40 successful searches, reranks, and JSON single points, with no recorded failures/timeouts. All 40 searches retained one unique minimum, rather than ten. Thus completion is distinct from ensemble diversity, and the result does not prove that each molecule has only one physically accessible conformation. Electronic calls reportedly converged in 11–14 SCF iterations (median 12). Example substrate 1a logs report dielectric constant 4.3300, electronic temperature 300 K, maximum 250 SCF cycles, 10⁻⁶ Hartree SCF and 10⁻⁴ e wavefunction convergence; these example settings are not verified for every record.

Total recorded worker time was 13,725.8 s, with mean 343.1, median 340.4, minimum 122.8, and maximum 986.4 s per substrate. This is summed worker time, not wall-clock duration or an end-to-end comparison against UFF.

S5.3 | Population weights and descriptor definitions

Reranked energies were converted using $\Delta E_i = 627.509474(E_i - E_{\min})$ and

$$p_i = \frac{\exp[-\Delta E_i/(RT)]}{\sum_j \exp[-\Delta E_j/(RT)]}, \quad N_{\text{eff}} = \frac{1}{\sum_i p_i^2}, \quad (\text{S7})$$

with $R = 0.00198720425864083$ kcal mol⁻¹ K⁻¹. Reaction temperatures were 60 °C for 20 records, 80 °C for 16, and 100 °C for four, distinct from the logged internal ALPB temperature of 298.15 K. Since each ensemble contains one minimum, $p_1 = N_{\text{eff}} = 1$ and all constant population/energy-variance quantities were excluded from fitting. These are total-energy weights, not calculated free-energy populations.

For centred heavy-atom coordinates X , $Q = X^T X / N_h$ has eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$ and sum L . The ten geometry features are total atom count, heavy-atom count, $\sqrt{L}$, maximum heavy-atom span, the three eigenvalues, asphericity $\lambda_3 - (\lambda_1 + \lambda_2)/2$, planarity fraction λ_1/L , and elongation fraction λ_3/L . Fractions are zero when $L = 0$.

The 13 fitted electronic features are total energy per atom, total energy per electron, electronic energy per electron, total-minus-electronic energy per atom, HOMO, LUMO, HOMO–LUMO gap, molecular dipole norm, charge standard deviation, mean absolute charge, charge range, mean atomic-dipole norm, and maximum atomic-dipole norm. HOMO/LUMO use fractional occupations $> 0.1/\leq 0.1$; orbital energies and the gap are in eV. Element-specific counts and charge extrema for N, O, S, F, Cl, Br, and I form 28 additional audit fields, not fitted features. No catalyst-induced polarization or catalyst–substrate interaction energy is produced.

TABLE S6 | Nested-family CREST ablations, same 38 Fe(P7)Cl substrates. Dimensions are before fold-local filtering.

Representation	Columns	MAE	RMSE	R^2
Locked geometry	248	8.498	11.357	0.476
+CREST geometry	258	8.858	11.578	0.455
+xTB electronic	261	8.610	11.745	0.440
+Both blocks	271	8.876	11.860	0.429

TABLE S7 | Reported paired-bootstrap MAE improvement intervals (baseline minus augmentation). Positive values favour augmentation.

Augmentation	95% percentile interval (ee points)
CREST geometry	[−0.733, −0.009]
xTB electronic	[−0.691, 0.423]
Combined	[−1.000, 0.212]

The unrounded combined improvement is reported as −0.377 points, with 0.107 of bootstrap draws above zero. Subtracting the displayed rounded MAEs gives −0.378; the rounding difference is retained rather than silently corrected. No addition improves the locked reference on MAE, RMSE, or R^2 . No corresponding CREST yield experiment was reported.

S6 | Retrospective PULSE-Acquire experiments

S6.1 | Reveal protocol and acquisition scores

Independent progression begins from a central-plus-diverse two-substrate seed, then reveals one recorded label at a time. The initial seed is held fixed within deterministic route/estimator comparisons. At each prefix, feature processing and model fitting use only revealed labels; predictions and selection quantities are recorded before the next label is exposed. After revealing the selected substrate, the model is refitted and evaluated on the remaining unrevealed pool. These post-refit pool metrics and the pre-reveal selected-substrate error are distinct.

Let U, D, C, P denote the documented uncertainty, molecular-diversity, family-coverage, and predicted-performance score components. The frozen weighted scores are

$$S_{UD} = 0.40U + 0.40D + 0.20C, \quad (S8)$$

$$S_{\text{performance}} = 0.50P + 0.30U + 0.20D. \quad (S9)$$

These weights were specified in the study; literature on active learning and reaction optimization does not establish them [29–32]. The reported-order route follows source presentation order, which is not verified as the original chronological experimental order. Diversity-first prioritizes Morgan/Tanimoto distance and family coverage. Random routes sample whole substrates without replacement: 500 replicates are reported for Ridge and Tanimoto- k NN, no random Elastic Net summary is included in this comparison.

The release inventory identifies `algorithms/ALGORITHM_SPECIFICATIONS.md` and linked source code for the exact U, D, C, P definitions, normalization, tie-breaking, candidate-set conventions, and random-seed construction. Those details cannot be inferred solely from the weighted equations. In particular, the recorded family-coverage component is not necessarily a binary new-family indicator. Independent replay, source-assisted transfer, and the older multi-endpoint placement score are different algorithms; the latter is not silently adopted as PULSE-Acquire.

The documented Fe(P7)Cl Ridge example starts with 1a (92% ee) and 1am (45% ee). Before revealing 1af, its recorded components are $U = 0.9196$, $D = 1.000$, and $C = 0.500$, giving

$$S_{UD}(1af) = 0.4(0.9196) + 0.4(1.000) + 0.2(0.500) = 0.86784. \quad (S10)$$

The selected record then reveals 61% ee. This checks the displayed score arithmetic, not that every alternative scored lower; a full candidate event log is required for the latter claim.

S6.2 | Learning metrics and the separate dedicated comparison

For prefix sizes $n_1, \dots, n_m$, mean-prefix MAE is the arithmetic mean of the corresponding pool errors. The documented independent-trajectory convention is trapezoidal integration,

$$\text{AUC}_{\text{MAE}} = \sum_{j=1}^{m-1} (n_{j+1} - n_j) \frac{\tilde{M}(n_j) + \tilde{M}(n_{j+1})}{2}, \quad (S11)$$

where the documented specification describes $\tilde{M}$ as median learning-curve MAE. Replicate aggregation and prefix membership are specified by the trajectory manifests. AUC has units of percentage points times label count and is not mean-prefix error; absolute AUCs across unequal prefix ranges should not be ranked as comparable datasets.

The study-defined n_{90} is the first evaluated prefix satisfying $M(n) \leq M_{\text{full}} + 0.1(M_0 - M_{\text{full}})$, where M_0 is the intercept-only baseline and M_{full} the documented full-pool leave-one-out reference. It is not a claim of 90% prediction accuracy or sustained convergence. The original baseline/reference prediction rows and threshold calculations are needed to verify the reported values; the reference-MAE summary column alone is insufficient.

The relative reductions from reported order to uncertainty-diversity are 55.3% and 33.2%, rounded to one decimal place. They support retrospective learning efficiency at matched label budgets, not a demonstrated reduction in future laboratory experiments. The dedicated record also stores “AUC” values of 780.199/348.624 for Fe(P7)Cl and 315.717/210.884 for Fe(CMC-Por)Cl (reported order/uncertainty-diversity). These behave as mean-prefix MAE times prefix count to rounding precision, not demonstrably as the independent experiment’s trapezoidal integral. They are therefore not mixed with the following AUC tables. The exact feature/split snapshot for the dedicated comparison requires its own manifest.

TABLE S8 | Dedicated matched Ridge route comparison: mean-prefix ee MAE. This is not the independent multi-estimator experiment.

Acquisition route	Fe(P7)Cl	Fe(CMC-Por)Cl aryl
Reported order	21.672	13.155
Diversity-first	11.197	9.842
Random	11.261	13.654
Performance-first	11.716	14.514
Uncertainty-diversity	9.684	8.787

S6.3 | Complete independent-progression summaries

Tables S9–S12 present all 56 documented independent route/estimator rows. The representation is fixed at `rdkit_morgan_stage2p`: 15 RDKit descriptors, 1,024 Morgan bits, and 248 Stage 2p fields (1,287 columns, P2 composition). This does not imply that Tanimoto-*k*NN uses non-fingerprint columns. It also does not convert the feature-arm results in Section S7 into independent-progression ablations.

Route codes D, H, P, R, and U mean diversity-first, reported order, performance-first, random, and uncertainty-diversity. Model codes EN, R, and K mean Elastic Net, Ridge, and Tanimoto-*k*NN. NA means the threshold was not reached. Final *n* is the number of revealed labels, not the number of test predictions. The reference MAE and Δ (final minus reference) are documented summary values; discrepancies of 0.001 from subtraction of rounded MAEs are retained. Final-prefix MAEs concern route-dependent remaining substrates and cannot establish superiority on a common test set.

TABLE S9 | Fe(P7)Cl enantioselectivity: independent-progression summaries.

Route	Model	Final <i>n</i>	Final MAE	AUC-MAE	<i>n</i> ₉₀	Ref. MAE	Δ
D	EN	37	4.454	343.044	4	13.238	-8.785
D	R	37	3.772	391.662	24	13.238	-9.466
D	K	37	1.215	368.284	19	13.238	-12.024
H	EN	37	8.838	741.152	NA	13.238	-4.400
H	R	37	9.719	757.662	NA	13.238	-3.519
H	K	37	3.969	601.044	37	13.238	-9.269
P	EN	37	16.511	403.781	NA	13.238	3.273
P	R	37	18.224	403.093	7	13.238	4.986
P	K	37	27.543	525.131	NA	13.238	14.304
R	R	37	8.122	355.527	29	13.238	-5.116
R	K	37	8.344	337.493	27	13.238	-4.894
U	EN	37	4.454	325.119	16	13.238	-8.785
U	R	37	3.772	337.181	16	13.238	-9.466
U	K	37	1.215	306.116	17	13.238	-12.024

Uncertainty-diversity minimizes ee AUC-MAE for all three Fe(P7)Cl estimators and for Fe(CMC-Por)Cl Elastic Net/Ridge, but diversity-first gives lower Fe(CMC-Por)Cl Tanimoto-*k*NN AUC (171.515 versus 175.443). The latter estimator also reaches *n*₉₀ earlier under reported order than under uncertainty-diversity (7 versus 14 labels). Thus threshold convergence and integrated error can disagree.

Yield is not a co-primary composite objective. Fe(CMC-Por)Cl reported-order final-prefix yield errors of 8.173 and 10.244 for Elastic Net/Ridge are much smaller than 35.582 and 35.032 under uncertainty-diversity, despite smaller differences in integrated errors. Fe(P7)Cl uncertainty-diversity yield also depends on estimator. These exceptions must remain visible

TABLE S10 | Fe(P7)Cl isolated yield: independent-progression summaries.

Route	Model	Final n	Final MAE	AUC-MAE	n_{90}	Ref. MAE	Δ
D	EN	38	15.999	561.248	2	14.742	1.258
D	R	38	16.006	570.357	2	14.742	1.265
D	K	38	0.509	469.261	12	14.742	-14.232
H	EN	38	22.180	908.899	2	14.742	7.438
H	R	38	23.917	913.166	2	14.742	9.176
H	K	38	16.019	633.651	37	14.742	1.278
P	EN	38	15.999	657.697	2	14.742	1.258
P	R	38	16.006	666.479	2	14.742	1.265
P	K	38	0.509	513.326	15	14.742	-14.232
R	R	38	18.447	639.424	4	14.742	3.705
R	K	38	13.461	484.752	26	14.742	-1.281
U	EN	38	15.999	547.182	2	14.742	1.258
U	R	38	16.007	554.164	2	14.742	1.265
U	K	38	0.509	452.942	15	14.742	-14.232

TABLE S11 | Fe(CMC-Por)Cl enantioselectivity: independent-progression summaries.

Route	Model	Final n	Final MAE	AUC-MAE	n_{90}	Ref. MAE	Δ
D	EN	25	4.111	217.781	13	11.580	-7.469
D	R	25	4.141	221.203	14	11.580	-7.439
D	K	25	2.089	171.515	13	11.580	-9.490
H	EN	25	5.352	292.463	12	11.580	-6.228
H	R	25	8.773	297.250	16	11.580	-2.807
H	K	25	1.482	238.721	7	11.580	-10.098
P	EN	25	5.352	336.989	23	11.580	-6.228
P	R	25	8.773	331.021	23	11.580	-2.807
P	K	25	1.482	187.666	6	11.580	-10.098
R	R	25	10.644	274.596	19	11.580	-0.935
R	K	25	9.781	225.759	12	11.580	-1.799
U	EN	25	4.111	193.940	6	11.580	-7.469
U	R	25	4.141	195.890	8	11.580	-7.439
U	K	25	2.089	175.443	14	11.580	-9.490

when describing the ee acquisition result. Unreported failures are unavailable, and no route reproduces the full historical decision process.

TABLE S12 | Fe(CMC-Por)Cl isolated yield: independent-progression summaries.

Route	Model	Final n	Final MAE	AUC-MAE	n_{90}	Ref. MAE	Δ
D	EN	25	35.582	453.058	6	15.000	20.582
D	R	25	35.032	467.224	6	15.000	20.032
D	K	25	26.904	354.284	2	15.000	11.904
H	EN	25	8.173	457.616	6	15.000	-6.827
H	R	25	10.244	474.511	6	15.000	-4.756
H	K	25	3.045	398.990	4	15.000	-11.955
P	EN	25	8.173	613.981	9	15.000	-6.827
P	R	25	10.244	616.732	9	15.000	-4.756
P	K	25	6.276	312.254	2	15.000	-8.724
R	R	25	21.604	518.475	24	15.000	6.604
R	K	25	18.131	390.508	2	15.000	3.131
U	EN	25	35.582	467.958	6	15.000	20.582
U	R	25	35.032	479.375	6	15.000	20.032
U	K	25	26.904	368.317	2	15.000	11.904

S7 | Guarded reciprocal PULSE-Transfer

S7.1 | Experts, suppression, and acceptance criteria

Transfer is evaluated in both source-to-target directions, excluding the 13 canonical shared-compound counterparts from source training. Target-only, source-only, pooled/domain-aware, source-regularized, residual-correction, and guarded predictions are conceptually distinct in the documented specification. Before each target label is revealed, eligible experts predict it; only afterward are target-domain cumulative errors updated. The primary replay estimator is standardized Ridge, $\alpha = 10$, bounded to $[0, 100]$. Ee and yield retain separate errors, weights, and metrics.

The target expert starts with weight 0.5. Source-informed experts share the remaining 0.5, and their total contribution is capped at 0.5 during the first five scored post-seed reveals. The implementation documentation describes GuardedEnsemble suppression using a greater-than-one-ee-percentage-point average-error condition; suppressed experts retain shadow predictions. The transfer contract also lists a 10% relative family threshold, which is not equivalent to the documented absolute-error suppression condition. Enforcement of this additional contract threshold is not established for the reported runs. Accordingly, the method description attributes suppression only to the documented absolute-error rule and does not claim implementation of the relative family rule. The initialization specification refers to label-independent coverage and reciprocal grouped-transfer reliability; absence of held-out target-label influence depends on the initialization provenance. The relevant specification, contract, and prediction records are indexed in Sections S9.3 and S9.2.

The prespecified acceptance criterion requires at least 10% lower early-prefix AUC-MAE, preservation or improvement of the full trajectory, no material family/scaffold-specific negative transfer, superiority to matched controls, and replication in both directions. Directional MAE improvements alone do not satisfy this conjunction. The reported evidence does not establish the full criterion.

S7.2 | Complete fixed-replay feature-arm comparisons

Tables S13 and S14 reproduce all ten feature arms in both directions. T and G denote target-only and guarded MAE, and $\Delta = G - T$, so negative values favour guarding. The AUC column is target-only AUC associated with that arm, not guarded AUC or the randomized-route mean. The tables support the connectivity/compact-pose comparisons in main-text Results 3.1 as well as the fixed-replay transfer statements in Results 3.3. They do not provide the absolute guarded early-prefix AUC needed to test the complete gate.

TABLE S13 | Reciprocal ee feature-arm comparison: Fe(CMC-Por)Cl $\rightarrow$ Fe(P7)Cl.

Arm	Columns	T MAE	T AUC-MAE	G MAE	Δ
B1	6	12.895	470.723	12.802	-0.092
B2	15	12.714	466.054	12.529	-0.185
B3	143	11.941	436.753	11.612	-0.329
B4	271	12.165	435.505	11.964	-0.201
B5	1039	11.815	424.292	11.654	-0.162
P0	248	19.317	695.948	16.528	-2.789
P1	254	19.355	697.280	16.425	-2.930
P2	1287	16.953	606.210	14.457	-2.496
P3	1083	11.276	397.809	10.973	-0.303
P4	1099	11.061	390.619	10.741	-0.320

B5 has the lowest target-only MAE among the two-dimensional arms in both domains, but B3 has a slightly lower guarded 2D MAE toward Fe(P7)Cl (11.612 versus 11.654). P4 has the lowest guarded MAE toward Fe(P7)Cl and P3 toward Fe(CMC-Por)Cl. The large reductions for poor pose-only baselines do not mean that these baselines outperform molecular connectivity. Counts and analysis-specific population mappings are described in Section S2 and identified by the accompanying ID lists.

TABLE S14 | Reciprocal ee feature-arm comparison: Fe(P7)Cl → Fe(CMC-Por)Cl.

Arm	Columns	T MAE	T AUC-MAE	G MAE	Δ
B1	6	14.243	430.423	13.053	-1.190
B2	15	14.100	426.091	12.955	-1.144
B3	143	12.563	379.429	12.083	-0.480
B4	271	12.578	378.166	11.781	-0.797
B5	1039	12.234	372.493	11.702	-0.532
P0	248	18.326	551.216	15.578	-2.748
P1	254	20.412	612.226	18.015	-2.397
P2	1287	17.659	533.867	14.239	-3.420
P3	1083	12.205	373.801	11.480	-0.725
P4	1099	12.208	373.214	11.509	-0.699

S7.3 | Randomized routes, scaffold exclusion, and matched controls

TABLE S15 | Reported 500-route paired-bootstrap changes in ee MAE, guarded minus target-only. Directions are source to target. These are route-aggregated estimates, not differences of the fixed-replay rows.

Source → target	Arm	Difference	95% interval
CMC-Por → P7	B5	−0.230	[−0.254, −0.205]
P7 → CMC-Por	B5	−0.399	[−0.443, −0.356]
CMC-Por → P7	P4	−0.417	[−0.451, −0.382]
P7 → CMC-Por	P4	−0.501	[−0.533, −0.466]

The documented target/guarded randomized-route AUC values are 458.185/451.743 and 397.759/386.411 for B5 (toward P7 and CMC-Por, respectively). Corresponding P3 values are 442.672/432.871 and 385.327/373.060; P4 values are 440.725/426.875 and 393.896/380.428. These are not the dedicated-acquisition stored AUC fields. P3 route MAE effects are reported as negative in both directions. The release inventory identifies row-level route records and bootstrap summaries in `prediction-validation/transfer-and-route-artifacts/`; aggregate values alone do not determine the corresponding paired interval.

TABLE S16 | Scaffold-held-out B5 replay. Aggregate MAEs and scaffold-bootstrap effects have different averaging units and must not be equated.

Quantity	P7 → CMC-Por	CMC-Por → P7
Target-only MAE	24.103	14.816
Guarded MAE	18.382	10.643
Scaffold groups / predictions	12 / 636	22 / 1,202
Scaffold-bootstrap G–T	−2.208	−4.596
95% interval	[−4.567, −0.644]	[−7.848, −1.888]
Incremental P3–B5 guarded MAE	−1.826	0.666
95% interval for P3–B5	[−3.968, 0.272]	[−0.589, 1.856]

The P3 incremental intervals include zero in both scaffold directions. In particular, the scaffold-bootstrap effect is not obtained by simply subtracting the displayed aggregate guarded and target MAEs. The underlying group weighting, resampling seed, and prediction rows are required to reproduce these estimates.

TABLE S17 | Reported guarded ee MAEs for pose controls. The two columns are source-to-target directions. Random and permuted controls address different hypotheses.

Representation/control	CMC-Por → P7	P7 → CMC-Por
P3 compact pose	10.973	11.480
P3 substrate-permuted pose	11.623	12.373
P3 random-feature control	11.891	13.357
P2 complete pose block	14.457	14.239
P2 substrate-permuted pose	13.793	17.289
P2 random-feature control	12.162	14.468

The complete P2 block does not consistently outperform its controls. For the secondary yield endpoint, B5 target/guarded MAEs are 13.762/13.537 toward Fe(P7)Cl and 16.194/15.675 toward Fe(CMC-Por)Cl. P3/P4 guarded yield errors remain above B5 in both directions according to the archive. A yield improvement cannot substitute for passing the ee transfer criterion.

S8 | PULSE-Explore: candidate generation, enumeration, and proposal limits

S8.1 | Production record and filtering contract

The documented recurrent production generator is SMILES-RNN [33] with a ChEMBL28 prior, recorded commit c8ee705961b4411707c69f73d309c1cf61208b95, and prior checkpoint ChEMBL28pur.ckpt. PromptSMILES [34] version 1.7.2 is separately recorded; interface smoke tests are not treated as production-generation evidence. Five seeds, 1101, 2202, 3303, 4404, and 5505, each generated 10,112 strings, giving 50,560 raw records. All 560 above the originally requested 50,000 were retained. A deterministic comparator with a reported 38-fragment vocabulary used the same raw budget. The release inventory identifies generation configuration, vocabulary, streams, checkpoint provenance, and hashes under candidate-generation/ppt1/. Recurrent molecular generation has precedent in the cited work, but that precedent does not validate the present candidate pool [35].

The stated screen proceeds through parsing/sanitization, canonical deduplication, aryl-azide identity, grammar membership, unique four-bond target assignment, sanitizable five-member closure, functional-group/azide filters, novelty/applicability, and target-definition checks. The release specification defines the applicability condition as maximum Morgan Tanimoto similarity ≤ 0.35 . Reference membership and the calculation of this statistic are specified through the frozen configuration and algorithm document; this similarity inequality is not interpreted as a distance threshold. The threshold must not be interpreted as a probability of synthetic success. A computational filter does not establish procurement, route feasibility, safety, or completion of chemist/environmental-health-and-safety review.

TABLE S18 | Reported generation funnel. Counts are molecular-string or structure populations, not independent experiments.

Population	Recurrent generator	Enumeration
Raw strings	50,560	50,560
Canonical unique structures	9,184	25,955
Structural-contract compliant	563	18,545
Within applicability domain	483	18,350
Scored by frozen predictor	483	18,223

S8.2 | Diversity and prediction-only comparison

For the reported matched 500-candidate samples, mean pairwise Tanimoto similarities are 0.312 (generator) and 0.412 (enumeration); corresponding mean maximum similarities to the 91-reference collection are 0.487 and 0.457. Mean distances/novelty proxies are consequently 0.688/0.588 internally and 0.513/0.543 relative to reference. The diversity analysis uses a diagnostic stratified screening sample, configured by stratified_screening_pool_per_provenance=500, not the final 483-row contract-valid/in-domain generated pool. These diversity statistics characterize screening samples, not a matched comparison of the final in-domain scored populations. The summary values do not prove greater extrapolation from the training domain.

The enumeration scoring artifact contains only the first 50,000 raw rows. IDs enum-050000–enum-050559 were omitted: 127 were structurally valid and in-domain, six structurally valid but out-of-domain, and 427 invalid/duplicate. Thus 18,350 – 127 = 18,223 eligible records were scored. This is a file-truncation omission, not chemical rejection. Restoring scores requires a new versioned comparison rather than silently changing the frozen panel or means.

S8.3 | Morphology and the prospective panel

The 483 in-domain generator candidates are reported as 192 heteroatom-rich, 117 flexible sp³-rich, 113 fused/polycyclic, 34 diaryl/bicyclic, and 27 simple aryl-tether structures. These sum to 483. A 16-member morphology-balanced intermediate set is distinguished from the final sealed 16-substrate proposal panel. The final panel comprises eight PULSE-Cat-branch and eight comparator candidates, four in proposed round A and four in round B for each branch. Archived records state blank experimental ee and outcome_revealed=false.

TABLE S19 | Frozen Morgan–Ridge predicted ee statistics for the scored populations only. None is an experimentally measured outcome.

Statistic (% ee)	Generator (<i>n</i> = 483)	Enumeration (<i>n</i> = 18, 223)
Mean	56.017	75.305
Standard deviation	7.459	6.569
Median	55.637	75.824
90th percentile	64.934	83.280
Maximum	83.302	96.452
Top-ten mean	69.411	85.806

The release inventory indexes the development shortlist and separate sealed 8+8 panel in the candidate-generation tree, including structures, selection records, round assignments, and lock hashes (Section S9.5). The panel is prediction-only; no experimental success or improved scope expansion is inferred from its selection.

S9 | Computational supporting data and reproducibility

The computational records will be supplied in SI.zip as a separate supplementary file accompanying the manuscript submission. The package organization below follows the release inventory dated 9 September 2026. All paths in this section are relative to the extracted archive; the files are not embedded in this PDF. The inventory distinguishes the experimental source records, retrospective model evaluations, and prediction-only candidate populations; these categories do not represent interchangeable observations.

S9.1 | Curated substrate datasets

- `curated-datasets/p7-records.jsonl` and `curated-datasets/cmcpor-records.jsonl` are the designated ledgers for the 41 retained Fe(P7)Cl and 50 Fe(CMC-Por)Cl records, respectively. The latter comprise 33 aryl substrates and 17 sulfonyl controls. Records include substrate IDs, starting-material SMILES, experimental ee/yield, reaction information, and literature provenance.
- The `curated-datasets/` directory indexes starting-material tables, candidate sites, atom mappings, reaction-centre decisions, and validation records. Family/scaffold assignments, shared identities, and exclusion reasons are associated with the relevant ledgers and analysis-specific splits. `dataset-id-lists.csv` identifies membership of the distinct development, CREST, progression, and transfer cohorts summarized in Table S2.
- `curated-datasets/p7-optimization-conditions.csv` contains the P7 reaction conditions and 16-condition optimization table. Source fields and linked source-ledger/SMILES tables identify document and page/table locations for the reported experimental outcomes.
- Twenty-nine P7 records lack curated product mapping. Their starting structures and experimental labels are retained, while product fields remain null. Exclusions of 1z, 1ad, and 1an have distinct reasons; exclusion is not equivalent to zero ee, and a measured zero-ee outcome remains valid.

S9.2 | Predictions, validation splits, and feature matrices

- `prediction-validation/p7-model-reports/` is the designated location for observed values, row-level structure-only predictions, residuals, split IDs, validation MAEs, and selected estimator parameters. Split files include `full-scope-structure-splits.json`, `family-heldout-ee-splits.json`, and their yield equivalents.
- `prediction-validation/p7-stage2/features/` indexes structure, pose, and yield matrices. Ordered columns and substrate IDs distinguish the development matrices from the reciprocal representations; dimensions alone do not identify a feature set.
- `prediction-validation/transfer-and-route-artifacts/` indexes reciprocal B5/P3/P4 matrices, 500-route records, scaffold replays, and bootstrap summaries. The row-level prediction and route records are the bootstrap inputs. Replicate counts, seeds, and calculation procedures are described in `code/scripts/ppt1/` and the algorithm specification.
- Recalculation of MAE, RMSE, R^2 , confidence intervals, and learning-curve statistics requires matching the endpoint, validation protocol, evaluation population, and resampling unit. Final-prefix errors from different acquisition routes are not common-test-set comparisons.

S9.3 | Acquisition, transfer, and metric definitions

`algorithms/ALGORITHM_SPECIFICATIONS.md` is the designated implementation specification for uncertainty, novelty/diversity, chemical-family coverage, predicted-performance scores, normalization, nearest-neighbor use, initial-substrate selection, tie-breaking, and candidate screening. It also documents transfer initialization and updates, error metrics, bootstrap intervals, learning-curve AUC, and n_{90} . The machine-readable transfer contract is `algorithms/ppt1-contract.json`.

The documented GuardedEnsemble suppression rule uses a greater-than-one-ee-percentage-point average-error condition, with shadow predictions retained for suppressed experts. The contract separately lists a 10% relative family threshold. These conditions are not interchangeable: the implementation documentation supports the absolute-error rule, whereas enforcement of the additional relative family threshold is not established for the reported runs. Section S7 therefore

reports directional transfer comparisons without claiming that both rules were implemented or that the complete acceptance criterion was met. The contract and implementation records are retained as distinct provenance documents; this clarification does not alter the algorithm or numerical results.

S9.4 | Structures, poses, occupancy fields, and calculation logs

- `geometry-calculations/pose/` indexes fixed catalyst references, constraints, retained Stage 1 poses, acceptance reports, and assemblies, together with Stage 2/2p/2pp feature JSONL files and manifests.
- `geometry-calculations/p7-density/` and `geometry-calculations/cmcpor-density/` index the 91 substrate-level occupancy representations and corresponding SVG views. The 13 shared-substrate overlays are indexed in `procedure/source-tables/si_pose_overlay_index.csv`. These are catalyst-domain-specific records, not 91 unique cross-domain chemical identities.
- CREST/xTB structures, outputs, and logs are indexed under `geometry-calculations/crest/` and `geometry-calculations/xtb/`. Feature definitions are recorded in the Stage 2p/Stage 2pp manifests, `procedure/si2-data-dictionary.md`, and the feature-generation source.
- The substrate-only CREST branch has one completed conformer per completed substrate and incomplete CMC-Por coverage. It is not a complete 50-substrate CMC-Por ensemble study. The P7 calculations and matched 38-substrate comparison are described separately in Section S5.2.
- Occupancy fields describe sampled geometry, not quantum electron density, Boltzmann populations, or selectivity-determining transition structures.

S9.5 | Candidate generation, screening, and proposal panels

- `candidate-generation/ppt1/` indexes generator configuration, SMILES-RNN checkpoint provenance, seeds, raw streams, enumeration vocabulary, rejection records, and hashes. Table S18 distinguishes the raw, canonical, structurally eligible, in-domain, and scored populations.
- The release specification defines the applicability condition as maximum Morgan Tanimoto similarity ≤ 0.35 . The reference set and statistic are specified through the frozen configuration and algorithm document. This condition is a computational screening definition, not a probability of experimental success; the reported inequality is not interpreted as a distance threshold.
- The diagnostic diversity comparison uses `stratified_screening_pool_per_provenance=500`. These stratified screening samples are distinct from the final 483-row contract-valid/in-domain generated pool and the final scored enumeration population.
- The candidate tree indexes the 16-substrate development shortlist and the separate sealed 8+8 prediction-only panel, including structures, round assignments, selection records, and lock hashes. No experimental outcome is inferred from a prediction or selection record.
- `candidate-generation/omitted-127-eligible-enumeration-candidates.csv` identifies the eligible enumeration records excluded from the frozen ee comparison. They correspond to `funnel-enumeration-ad.csv` IDs `enum-050000–enum-050559` with `contract_valid=true` and `ad_status=in_domain`. The ee table contains only the first 50,000 raw rows; these 127 candidates have no prediction rows and remain excluded from all scored-population statistics.
- Accordingly, Table S19 compares 483 generated candidates with 18,223 scored enumerated candidates, not all 18,350 eligible enumerated candidates. No missing prediction is imputed. Inclusion of the excluded records would constitute a new, versioned scoring analysis rather than the frozen comparison reported here.

S9.6 | Code, software records, and provenance

- `code/scripts/` and `code/feature-source/` index analysis/validation scripts and feature-generation source. Execution procedures are documented in `code/REPRODUCE.md` and the `procedure/` documents.

- Dependency declarations are `code/requirements.txt` and `code/requirements-generator.txt`. `code/SOFTWARE_VERSIONS.md` records runtime versions and external commit/checkpoint pins. These are dependency and provenance records, not complete environment lockfiles.
- File-level manifests are `procedure/si2-manifest.json`, `procedure/si3-manifest.json`, `procedure/si4-manifest.json`, and `procedure/compact-si-data-manifest.json`. Figures and plot data are indexed in `figures/` and `procedure/source-tables/`.
- `DATA_AVAILABILITY_STATEMENT.md` documents the computational package's distribution details. The companion `SI.zip` file is the supplementary delivery described in this manuscript.

S9.7 | Reproduction scope and archive use

For reproduction, extract the companion archive while preserving its directory structure and follow `code/REPRODUCE.md`. Required data and coordinate files must be included as file contents, not merely as links to the original staging repository; the manifests provide the file-level provenance. Dependency declarations, runtime-version records, and external pins document the reported software environment. The release inventory does not identify a complete environment lockfile or container digest, so exact environment reconstruction and bitwise-identical outputs are not established by those records alone.

Reproduction should retain each analysis's population IDs, endpoint, feature manifest, split or reveal order, and evaluation convention. The transfer comparisons do not verify enforcement of the additional contract-level family safeguard (Section S9.3). The reported counts, numerical comparisons, and explicit 127-record scoring exclusion are retained; diagnostic screening samples are not substituted for final prediction pools.

S9.8 | Data availability and study boundary

The curated datasets, feature matrices, prediction and split records, geometry/calculation outputs, candidate-screening records, and analysis code will be supplied in the companion `SI.zip` file at manuscript submission. Sections S9.1–S9.6 identify their locations within the archive; execution instructions and provenance are documented in `code/REPRODUCE.md`, `code/SOFTWARE_VERSIONS.md`, and the `procedure/` manifests. Experimental source data originate from the two cited reaction studies [1, 2]. The distribution route described here is the accompanying supplementary archive; no separate repository deposit or DOI is claimed.

This study concerns retrospective prediction, directional transfer diagnostics, geometric representations, and prediction-only candidate generation. No experimental outcomes are available for the proposed generated panel or the five P7 prediction-only substrates. Supplier evidence, route acceptance, safety review, and a closed-loop laboratory campaign are outside the present analysis. The results therefore do not establish reduced laboratory effort, experimentally successful scope expansion, or satisfaction of the complete transfer acceptance criterion.

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